

CO3-AT2 CASE STUDY

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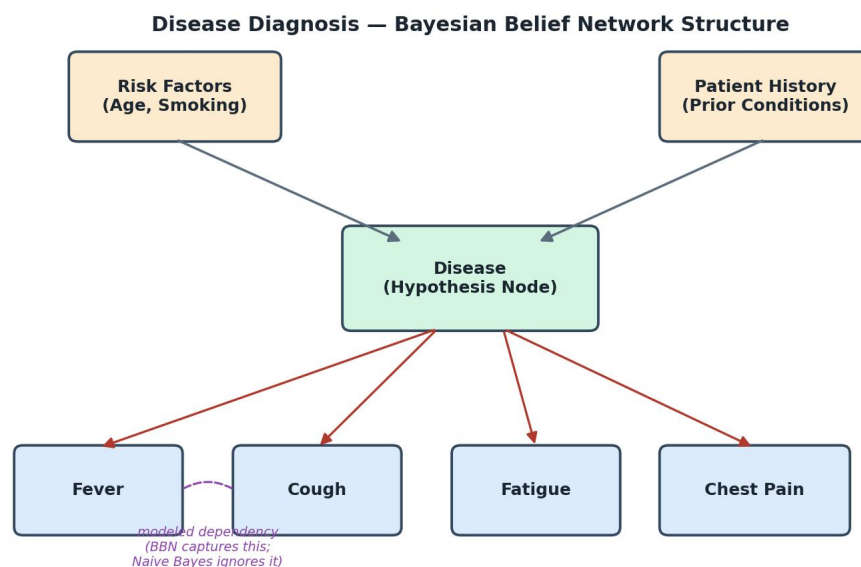
Course Name: MACHINE LEARNING

Case Study 1: Probabilistic Disease Diagnosis System

System Design

A layered probabilistic design combines three tools, each suited to a different level of the problem:

- Bayes' Theorem (single symptom): for a quick first-pass estimate, $P(\text{Disease} \mid \text{Symptom}) = \frac{P(\text{Symptom} \mid \text{Disease}) \cdot P(\text{Disease})}{P(\text{Symptom})}$, using known disease prevalence as the prior.
- Naïve Bayes classifier (many symptoms, fast screening): combines many symptoms at once, assuming each is conditionally independent given the disease — $P(\text{Disease} \mid S_1, \dots, S_n) \propto P(\text{Disease}) \cdot \prod P(S_i \mid \text{Disease})$. Useful as a fast triage layer over dozens of symptoms/lab values.
- Bayesian Belief Network (final diagnostic model): a Directed Acyclic Graph where Risk Factors and Patient History feed into a Disease node, which in turn influences multiple Symptom nodes — explicitly modelling that some symptoms (e.g., Fever and Cough) are themselves correlated, not just independently caused by the disease.



Handling Uncertainty

- Every diagnosis is expressed as a probability distribution over possible diseases, not a single verdict — appropriate for medicine, where certainty is rarely available.
- As each new symptom, test result, or history item is entered, the posterior is updated sequentially (today's posterior becomes tomorrow's prior), letting the system refine its belief as evidence accumulates rather than requiring all information upfront.
- Missing or unknown evidence is naturally handled by marginalizing (summing) over the unobserved variables, instead of requiring a value for every field.

Conditional Dependencies Between Symptoms

Naïve Bayes' independence assumption is often violated in medicine — e.g., fever and cough frequently co-occur beyond what the disease alone explains (shared underlying inflammation response). A BBN captures such dependencies explicitly through its graph structure and conditional probability tables, which generally improves diagnostic accuracy over Naïve Bayes, at the cost of needing more data and expert input to define the extra dependency edges correctly.

Role of Probability Learning

- Conditional probability tables (CPTs) are learned from historical patient records — simple counting/MLE when data is abundant, or Bayesian parameter estimation (with clinician-elicited priors) when data for a given disease is scarce, which stabilizes estimates and avoids overfitting to a handful of rare cases.
- Network structure itself can be refined over time (structure learning) as more data reveals previously unmodelled dependencies between symptoms.

Evaluation Against Real-World Constraints

Constraint	Effect / Mitigation
Rare-disease data scarcity	Sparse data → unstable MLE estimates; mitigate with smoothing and informative priors from clinical literature/experts.
Interpretability requirement	Clinicians need to trust and audit the reasoning — BBNs are naturally interpretable (explicit graph and probabilities) versus opaque black-box models.
Computational cost	Exact inference in large BBNs is NP-hard in general; large networks require approximate inference (sampling, variational methods) for real-time use.

Constraint	Effect / Mitigation
Regulatory/ethical constraints	Outputs must be calibrated, auditable, and free of demographic bias — requires ongoing validation against diverse patient populations before clinical deployment.

Overall, the layered approach is effective as a decision-support aid (ranking likely diagnoses, flagging urgent cases) but should augment, not replace, clinician judgement — particularly given data scarcity for rare conditions and the high cost of a missed diagnosis.

Case Study 2: Email Spam Detection System

System Design

- Feature extraction: tokenize email text, remove stop words, build a word-frequency/TF-IDF vector; additional features such as sender reputation, link count, and attachment presence.
- Naïve Bayes classifier: computes $P(\text{Spam} \mid \text{words}) \propto P(\text{Spam}) \cdot \prod P(\text{word}_i \mid \text{Spam})$ using word-frequency statistics — fast, scales to large vocabularies, works well as the primary classifier.
- Bayes Optimal Classifier: rather than committing to one trained model, combine predictions from several candidate models/feature-sets, each weighted by its posterior probability given the training data, to get the theoretically lowest-error decision achievable from the hypothesis space — used here as an ensembling layer over a few strong candidate Naïve Bayes variants (since true Bayes-optimal combination over all hypotheses is computationally infeasible).

MLE vs MDL in Feature Selection

Aspect	Maximum Likelihood (MLE)	Minimum Description Length (MDL)
Objective	Choose features/parameters that maximize the likelihood of the training labels.	Minimize $\text{description-length}(\text{model}) + \text{description-length}(\text{data} \mid \text{model})$ — fit AND simplicity.
Feature-set size tendency	Tends to keep adding features that improve training likelihood, risking overfitting with a large vocabulary.	Penalizes extra features/parameters, naturally producing a more compact, regularized feature set.
Generalization	Can overfit on noisy or high-dimensional word features, especially	Better generalization on unseen emails by explicitly trading off complexity

Aspect	Maximum Likelihood (MLE)	Minimum Description Length (MDL)
	with limited training emails.	against fit.

System Performance and Trade-offs

- Model complexity trade-off: a larger vocabulary/feature set can capture subtle spam indicators but increases variance and overfitting risk, especially against adversarial spammers who deliberately vary wording; MDL-based feature pruning mitigates this by discarding uninformative or overly rare tokens.
- Precision vs recall: a spam filter is usually tuned to favor precision (avoid blocking legitimate mail) even at some cost to recall, since a false positive (blocked real email) is typically more costly to the user than an occasional spam message getting through.

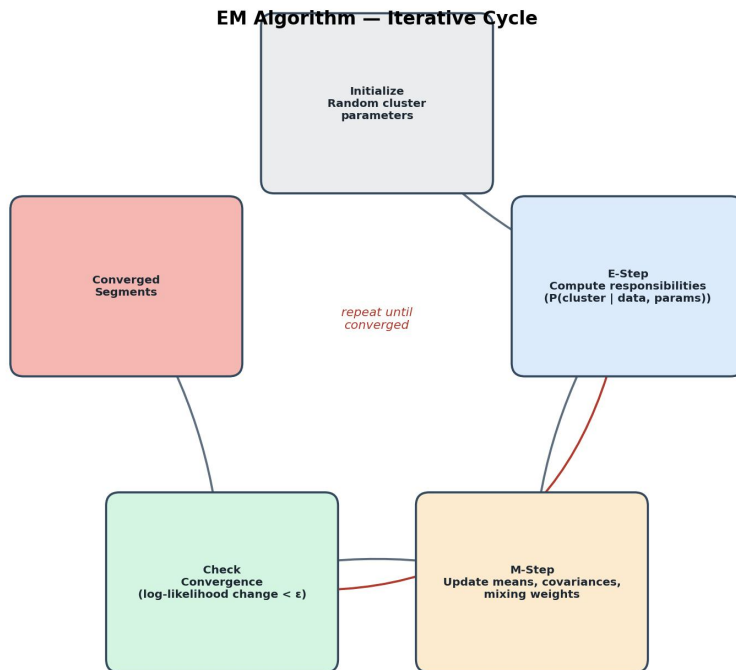
Effect of Sample Size and Hypothesis Space

- With a small training set and a large hypothesis space (many possible word-feature combinations), many different feature-weight hypotheses fit the training data equally well, but most generalize poorly — high variance, unstable accuracy on new emails.
- As sample size grows, the effective hypothesis space that remains consistent with the data narrows (analogous to version-space convergence in concept learning), and classification accuracy stabilizes and improves.
- MDL's built-in complexity penalty effectively shrinks the practical hypothesis space regardless of sample size, so it needs fewer training examples than an unconstrained MLE search to reach reliable generalization — a useful property when labeled spam data is limited.

Case Study 3: Customer Segmentation Using the EM Algorithm

Applying EM

Model each customer's purchasing behaviour (e.g., spend amount, purchase frequency, category mix) as generated by one of K hidden segments, each following a Gaussian distribution (a Gaussian Mixture Model). The segment label is the hidden/unobserved variable EM is designed to handle:



- E-step: given current cluster parameters, compute each customer's probability of belonging to each segment (soft "responsibility" values).
- M-step: given those responsibilities, re-estimate each segment's mean, covariance, and mixing weight to best explain the (softly) assigned customers.
- Repeat until the log-likelihood improvement between iterations falls below a small threshold ϵ .

Convergence Behaviour

EM guarantees the data log-likelihood never decreases from one iteration to the next, so the process reliably converges — but only to a local optimum, not necessarily the global best clustering. In practice, log-likelihood typically rises quickly in the first few iterations and then flattens.

Initialization Challenges

- Random initialization of cluster means can lead to poor local optima, empty or near-duplicate clusters, or slow convergence.
- Mitigations: run EM multiple times from different random initializations and keep the best result; seed initial means using K-means++ or hierarchical clustering rather than pure random placement.

Role of Probability Learning

Unlike hard clustering (e.g., K-Means), EM's soft cluster assignments reflect genuine ambiguity in customer behaviour — a customer near a segment boundary is assigned partial probability to multiple segments rather than forced into one, which is a more realistic model of real purchasing behaviour and supports more nuanced, probability-weighted marketing decisions.

Using Results for Targeted Marketing

- Segment profiles (e.g., "high-frequency low-spend", "occasional big-ticket buyer") derived from the fitted cluster parameters guide campaign design per segment.
- Soft assignments allow blended targeting — e.g., a customer 60% in "deal-seeker" and 40% in "loyal high-spender" can receive a mixed offer rather than being pigeonholed.

Limitations

- Assumes a parametric (typically Gaussian) shape for each segment, which may not match real, often skewed or multimodal, purchasing distributions.
- The number of segments K must be chosen in advance (e.g., via BIC/AIC comparison), and results are sensitive to that choice.
- Sensitive to outliers (a few extreme high-spenders can distort a cluster's estimated mean/covariance) and can converge only to a local — not global — optimum.
- Scalability to very large, high-dimensional retail datasets requires efficient implementations (e.g., mini-batch EM) and careful handling of missing/incomplete purchase records.

Case Study 4: Concept Learning for Medical Image Classification

System Design

Extract discrete, clinically meaningful attributes from each image (e.g., texture: Smooth/Irregular; shape: Round/Spiculated; size: Small/Large; margin: Well-defined/Ill-defined). Each attribute-conjunction over these discretized features defines a hypothesis (e.g., <Irregular, Spiculated, ?, Ill-defined> → Abnormal). Apply the Candidate-Elimination algorithm to maintain the version space — bounded by specific boundary S and general boundary G — consistent with all labeled (Normal/Abnormal) training images.

Finite vs Infinite Hypothesis Spaces

	Finite Hypothesis Space (discretized attributes)	Infinite Hypothesis Space (continuous/unrestricted)
Example	Conjunctions over k discrete attributes:	Arbitrary real-valued decision

	Finite Hypothesis Space (discretized attributes)	Infinite Hypothesis Space (continuous/unrestricted)
	$ H = 3^k + 1$.	thresholds, or unrestricted boolean/neural function classes.
Convergence guarantee	Candidate-Elimination provably converges to the target concept given enough noise-free, consistent examples.	No such finite guarantee in general; convergence and generalization instead governed by other complexity measures (e.g., VC-dimension).
Sample complexity	Grows with k but is analytically boundable (mistake-bound $\approx \log_2 H $).	PAC-style bounds instead scale with VC-dimension d : roughly $O((d/\epsilon) \cdot \log(1/\delta))$.

Sample Complexity and the Mistake Bound Model

- Finer attribute discretization or more attributes increases $|H|$ exponentially (3^{k+1}), which increases the number of examples needed for the version space to converge to a near-unique hypothesis — directly raising sample complexity.
- The mistake-bound model formalizes the worst-case number of misclassifications an online learner can make before converging; for a finite consistent hypothesis space, this is bounded by roughly $\log_2 |H|$ (as in the Halving algorithm) — giving a concrete, theory-grounded expectation of how many diagnostic errors the system might make while it is still learning.

Generalization to Unseen Data

- Candidate-Elimination only guarantees consistency with the training examples it has seen, not robustness to label noise: a single mislabeled training image can cause the true target concept to be prematurely excluded, or even collapse the version space to empty.
- Purely conjunctive hypotheses over hand-picked discrete attributes are often too restrictive to capture the full visual complexity of real medical images (subtle textures, overlapping structures) — in practice such systems are more useful as an interpretable baseline / teaching tool for the underlying learning theory, with production systems using richer hypothesis classes (e.g., ensembles or CNNs) while still relying on the same underlying sample-complexity and generalization principles.
- Recommended refinement for real deployment: combine the interpretable conjunctive concept space with a statistical/robust classifier (e.g., allowing a small tolerance for inconsistent examples) so a few noisy labels don't collapse the entire version space.

Case Study 5: Investment Decision-Making Under Market Uncertainty

System Design

- Represent market conditions as hidden regime hypotheses (e.g., Bullish / Bearish / Volatile), each with a prior probability informed by historical data and macroeconomic expert judgement.
- As new market data arrives (returns, volume, indicators), apply Bayesian inference to update the posterior over regimes: $\text{posterior} \propto \text{likelihood}(\text{new data} \mid \text{regime}) \times \text{prior}(\text{regime})$ — with today's posterior becoming tomorrow's prior for the next update (sequential/recursive Bayesian updating, similar in spirit to a Hidden Markov Model / Bayesian filter).
- When the space of candidate predictive strategies/models is too large to combine exactly (as a full Bayes Optimal Classifier would), use the Gibbs algorithm: randomly sample a single strategy with probability proportional to its posterior, and use it to make the current prediction — a computationally cheap approximation whose expected error is bounded at roughly twice the true Bayes-optimal error.

Influence of Prior Knowledge

- Strong, well-chosen priors (from domain expertise or long historical baselines) stabilize predictions when data for a given asset or period is sparse or noisy — important for new instruments or fast-moving events with little precedent.
- Conjugate priors (e.g., Beta for win/loss trade outcomes, Normal for returns) allow the posterior to be updated in closed form, keeping updates computationally cheap enough for frequent (even real-time) re-evaluation.
- Risk: an overly strong or stale prior can bias predictions and slow the system's adaptation when genuine regime shifts occur — priors must be periodically reviewed, not fixed permanently.

Evaluating Risk Using Posterior Probabilities

Because Bayesian inference produces a full posterior distribution over expected returns (not just a point estimate), standard risk measures — Value-at-Risk, probability of loss exceeding a threshold, credible intervals — can be computed directly from that distribution, letting decisions incorporate uncertainty and risk tolerance rather than relying on a single 'best guess' forecast.

Dynamic Adaptation to New Data

Sequential Bayesian updating lets the model revise its beliefs incrementally as each new trading period's data arrives, without retraining from scratch — supporting near-real-time adaptation to changing market conditions and enabling detection of regime shifts via monitoring posterior predictive checks (e.g., a run of data that is improbable under the current posterior signals a possible regime change).

Computational Challenges and Optimization Strategies

Challenge	Optimization Strategy
Exact Bayesian inference intractable in high-dimensional parameter/strategy spaces	Use approximate inference: Gibbs/MCMC sampling, or variational Bayes for faster approximate posteriors.
Real-time/low-latency trading constraints	Use conjugate-prior closed-form updates where possible; otherwise cheap sampling-based approximations (e.g., Gibbs algorithm) instead of full Bayes-optimal combination.
Non-stationary markets (violates i.i.d. assumptions)	Use hierarchical/adaptive priors and sliding-window or discounted updating so older data has less influence as conditions change.
Risk of overfitting to historical data / prior misspecification	Regularize via weakly-informative priors, cross-validate on out-of-sample periods, and periodically re-elicite priors from updated domain expertise.